PC Spec Checker - Beginner Guide (EN)

PC Spec Checker v3.0 — Complete Beginner's Guide (English)

> Made so that even people using a computer, smartphone, AI, or messenger app **for the very first time** can do everything by themselves with just this one document. If you see a word you don't know, check **18. Glossary** at the bottom.

> 🌐 한국어 버전: GUIDE.md

Table of Contents

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1. What is this? (one-line summary)

Double-click one file and it instantly shows you everything about your PC's specs — for free.

- No installation · No sign-up · Works without internet · Never sends your data anywhere
- It only **reads** your PC. It does not change or delete anything.

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2. Explained simply

Think of a **medical checkup** at a hospital. This tool is a **free health checkup for your computer**.

When you double-click it, the computer inspects its own parts and tells you:

- 🧠 What **CPU** (the brain) you have and how fast it is
- 💾 How much **RAM** (work space) you have
- 🗄 What **storage** (SSD/HDD) you have, how full it is, and whether it's healthy
- 🎮 What **graphics card (GPU)** you have
- 🔋 **Battery** health (if it's a laptop)
- 🛡 Security status such as **antivirus and firewall**
- 🧰 The **programs and developer tools** installed (auto-checks up to 193 tools)
- 🏆 Finally, a **score out of 100** rating how ready your PC is for development (coding)

> 💡 You don't need to care about "development" or "coding." It's also very useful just for checking **basic specs** like CPU, RAM, storage, and GPU.

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3. Who is this for?

If you are...	You can use it to...
Anyone curious about their PC specs	Check instantly without installing heavy software
Someone buying/selling a used PC	Summarize the specs on one page
Someone not familiar with computers	Just say "double-click this file" — done
A coding beginner	See if your PC is ready and what to install next

If you are...	You can use it to...
People coding with AI (ChatGPT, Claude, Cursor, etc.)	Verify your dev environment is set up
IT support staff	Tell a remote person "run this file and tell me what you see"

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4. Prerequisites / Required programs

You need to prepare almost nothing. Just confirm the following.

Item	Required?	Notes
Windows 10 or 11	✅ Required	This tool is **Windows only**.
PowerShell	✅ Required but **already installed**	Built into Windows. No separate install needed.
Internet connection	🔵 Optional	Works without it. With it, you also see an "internet response time" item.
Administrator rights	🔵 Optional	Works without them. Only a few items (like CPU temperature) show `N/A`.
Installing extra programs	❌ Not needed	You do **not** need to install antivirus, drivers, runtimes, etc.

> ❗ **It does NOT work on Mac or Linux.** Use it only on a Windows computer. Check your Windows version: press `Windows key + R` → type `winver` → Enter.

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5. How to download

This tool is just **one** file: `pc-spec_checker-kit.bat`. There are two ways to get it.

Method A — Get everything from GitHub (recommended)

1. In a web browser (Chrome, Edge, etc.), open the project's GitHub page: `https://github.com/sodam-ai/pc-spec_checker-kit`
2. Click the green `< > Code` button.
3. In the menu, click `Download ZIP` at the bottom.
4. A zip file like `pc-spec_checker-kit-main.zip` is usually saved to your `Downloads` folder.

Method B — Get just the one file

If you only received `pc-spec_checker-kit.bat`, that single file works completely on its own. (No other files are required.)

> 💡 Can't find the downloaded file? → See 15. File locations.

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6. How to install (nothing to install)

This tool has no "installation" step. Just extract it and use it.

1. Find the downloaded ZIP file. (Usually in the `Downloads` folder.)
2. **Right-click** the ZIP file → `Extract All` (or `압축 풀기`).
3. Inside the extracted folder you'll find `pc-spec_checker-kit.bat`.

> ✅ **Recommended location**: put it somewhere you can easily find, like your `Desktop` or `Documents`. ⚠️ **Avoid**: running it directly from inside the ZIP without extracting — it can behave unreliably. Always **extract first**, then run.

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7. Quick start (3 steps)

The fastest way to see results.

1. **Double-click** the `pc-spec_checker-kit.bat` file.
2. (If a security warning appears) click `More info` → `Run anyway`. → details in section 8
3. When the menu appears in the black window, press `1` and **Enter** → all specs appear at once.

> That's it! After viewing, press **Enter** to return to the menu, and press `0` to quit.

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8. How to run (in detail)

8-1. Basic run (double-click)

Double-click the `pc-spec_checker-kit.bat` file with your mouse. A black command window opens and shows the menu.

8-2. When Windows shows a security warning (this is normal)

The first time you run a `.bat` file downloaded from the internet, Windows may show a warning for safety.

- If a blue window saying **"Windows protected your PC"** appears:
  1. Click `More info`.
  2. Click the `Run anyway` button that appears below.

> 🔒 **Why is it safe?** This file is **100% open source**. Open `pc-spec_checker-kit.bat` with **Notepad** and you can read every line of code yourself. The tool only **reads** your PC info — it does not install, change, or send anything. (20. Safety)

8-3. Other ways to run

- You can also **right-click →** `Open` to run it.
- **Run as administrator**: right-click the file → `Run as administrator`. → This may reveal a few extra items (like CPU temperature) instead of `N/A`. (Not required.)

8-4. If you want to save the results to a file (optional)

This tool does not create files by itself. To save the results:

- **Option 1 (easiest)**: In the black window, drag-select the text and copy it (`Ctrl+C` or right-click), then paste (`Ctrl+V`) into Notepad.
- **Option 2 (run by command)**: see "Save results to a text file" in 14. Commands.

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9. How to use — every menu item 0–15

When you run the tool, the menu below appears. **Type the number of the item you want and press Enter.**

> If it's your first time, just choose `1` **(Full Scan)**.

No.	Menu name	What it shows
1	**Show ALL (Full Scan)** ⭐ recommended	Scans items 2–15 below **all at once**
2	Basic Info + Motherboard + BIOS	PC name, user, Windows edition/version/build, install date, uptime, Windows activation, latest update, motherboard maker/model, BIOS info
3	CPU (detailed)	Processor name/maker, core/thread count, clock speed, cache, 32/64-bit·ARM, virtualization support, **CPU temperature** (with admin), live usage bar
4	RAM (each stick)	Total/used/free memory, **per stick (slot)**: capacity, type (DDR4/DDR5, etc.), speed, maker, part number, slots used

No.	Menu name	What it shows
5	Disk (type + health)	Storage model/capacity, **SSD/HDD/NVMe type**, **health status** (Healthy/Warning/Unhealthy), disk temperature, per-drive (C:, D:) total/used/free bar
6	GPU (detailed)	Graphics card name/status, dedicated memory (VRAM), current resolution/refresh rate (Hz)/color depth, driver version/date, GPU temperature (if sensor present)
7	Network + Display	Network adapter info, **IP address**, MAC address, gateway, DNS, **internet response time** (one ping to 8.8.8.8), connected **monitor** name/maker
8	Battery / Power	(Laptop) battery name/status, **charge %**, time left, **current vs design capacity (battery health %)**, charge state / (Desktop) "No battery" + power plan name
9	Security Status	**Antivirus** name/active, **firewall** (Domain/Private/Public) on/off, **UAC** on/off, C: drive **BitLocker (encryption)** status
10	Startup Programs	Programs that **launch automatically when you turn on the PC** (registry, Startup folder, scheduled tasks), total count (too many can slow boot)
11	Audio / USB / Bluetooth	Sound devices, USB controllers, **connected USB devices**, Bluetooth devices
12	Installed Apps	Installed programs organized into **8 categories** (browsers, office, communication, media, graphics, cloud, security, utilities) + versions
13	**Dev Tools (deep scan)** 🟡	Checks **193 tools** in **17 categories** (Python, Node.js, Git, Docker, VS Code, AI coding tools, etc.). Shows versions if installed
14	WSL (Linux on Windows)	Whether WSL is installed, version, installed Linux distros, and whether key tools (python, node, etc.) exist inside Linux
15	**Score + Recommendations** 🏆	Rates your PC **out of 100** (grades S/A/B/C/D) + **weaknesses and recommendations**
0	**Exit**	Closes the tool

Usage flow

1. Type a number (e.g. `1`) → **Enter**
2. Results appear (a full scan can take 30–60 seconds)
3. When done, press **Enter** → back to the menu
4. Pick another item, or press `0` to quit

> 💡 One run lets you keep picking items. You don't need to double-click again each time.

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10. How it works (under the hood)

A simple explanation of how it **safely reads info only**.

1. **It only reads.** Using features already built into Windows (below), it simply **asks the computer "what are your specs?" and prints the answers** to the screen.
   - **WMI / CIM**: Windows' official store of its own hardware info. CPU, RAM, disk, etc. are queried from here.
   - **Registry (read-only)**: It **only reads** the list of installed programs, startup items, etc. It never edits them.
2. **It finds dev tools in 3 steps.** (So it can detect tools even if they're not on your PATH.)
   1. **PATH search** — finds the tool in registered paths and asks for its version
   2. **Known install paths** — checks dozens of folders where each tool is commonly installed (version-specific folders are found with wildcards)
   3. **Registry** — for programs with a window (GUI), it checks the installed list
3. **It touches the internet only once.** When you run item 7 (Network) or item 1 (Full Scan), it sends **one ping to Google's public address** `8.8.8.8` to check if the internet works. It **only checks for a response — it does not send your data**. With no internet, this item is simply skipped.
4. **It does not create files.** No temp files, logs, or reports are saved. All results show only in the window.
5. **It does not change settings.** No installing, deleting, or changing settings at all.

> Summary: **No install · No changes · No saving · No external transfer. Only read and show.**

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11. How to read the results

The meaning of the symbols and colors on screen.

Symbol	Meaning
`[O]` (green)	That tool/feature is **present / on / OK**
`[X]` (gray)	That tool is **not installed**
A number like `2.1.80`	The tool's **version**
`(found in PATH)` or `(found)`	The tool **exists, but its version couldn't be read**. It works fine.
`N/A`	Can't check right now. CPU temperature, etc. may need **administrator rights**.
`N/A (admin required)`	You may see it if you run as administrator.

Symbol	Meaning
Bar graph `[#######-----]`	Visual usage (%). Green = plenty, yellow = moderate, red = high

Color meaning (approximate)

- 🟢 **Green**: good / normal / plenty
- 🟡 **Yellow**: caution / moderate / worth checking
- 🔴 **Red**: risk / low / high temperature
- ⚪ **Gray**: not applicable / not installed / disabled

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12. The score system in detail

Menu 15 (or the end of the full scan in menu 1) rates your PC **out of 100**. The basis is "how suitable is it for development (coding)."

Points per category (100 total)

Category	Max	Scoring rule
CPU	20	8+ cores = 20 / 6 cores = 16 / 4 cores = 12 / 2 cores = 6 / fewer = 3
RAM	20	64GB+ = 20 / 32GB = 18 / 16GB = 14 / 8GB = 8 / less = 3
Storage	15	NVMe = 15 / regular SSD = 12 / HDD = 5 · (−3 if free space on C: is under 30GB)
GPU	10	Dedicated 8GB+ = 10 / 4GB = 8 / 2GB = 5 / integrated = 3 / none = 0
Dev Tools	35	See table below

Dev Tools 35-point breakdown

Group	Scoring	Max
Core (Python, Node.js, Git)	5 each	15
Package managers (npm, pnpm, pip)	2 each	6
Editors (VS Code, Cursor, IntelliJ, Neovim)	2 each	4
AI coding tools (Claude, Gemini, Codex, Aider)	2 each	4
Extra (Docker, GitHub CLI, Bun, Rust, Go, Make, Terraform, kubectl, AWS, ripgrep)	1 each	6

Grades

Score	Grade	Meaning
85–100	**S**	Excellent dev machine

Score	Grade	Meaning
70–84	**A**	Great setup
55–69	**B**	Decent, room to improve
40–54	**C**	Needs improvement
0–39	**D**	Consider upgrading hardware + tools

> Below the score, **weaknesses and recommendations** (e.g. "RAM under 16GB", "Python missing") are shown. ❗ A low score does **not** mean a "bad PC." This score is about **development readiness**. A perfectly good PC for web browsing/documents will score low if it has no dev tools.

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13. Workflow (full flow diagram)

```
[1] Download           Get the ZIP from GitHub  (or just the .bat file)
         │
         ▼
[2] Extract            Right-click the ZIP → "Extract All"
         │
         ▼
[3] Run                Double-click pc-spec_checker-kit.bat
         │
         ▼
[4] Handle warning     "More info" → "Run anyway"  (first time only)
         │
         ▼
[5] Pick a number      First time → 1 (Full Scan) → Enter
         │
         ▼
[6] View results       Read specs & score on screen (copy if needed)
         │
         ├─► See more items → Enter to return to menu → pick another number
         │
         ▼
[7] Exit               Type 0 → "Goodbye!" → window closes
```

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14. Commands

> Honestly, **you barely need to remember any commands.** Double-click with the mouse, then just press **numbers**.

The "inputs" used inside the tool (that's all there is)

Input	Action
`1` – `15` + Enter	Run that menu item
`1` + Enter	Full scan (most recommended)
`13` + Enter	Dev-tools deep scan only
`15` + Enter	Score only
Enter (on a result screen)	Back to the menu
`0` + Enter	Exit

(Optional) Run commands for people comfortable with the computer

You can also run it directly from PowerShell or Command Prompt (cmd). After navigating to the folder containing the file:

```
pc-spec_checker-kit.bat
```

Or with the full path:

```
"C:\path\to\your\folder\pc-spec_checker-kit.bat"
```

(Optional) Save results to a text file

In Command Prompt (cmd), this saves the results to a file. (Auto-enters `1` for a full scan and saves to `result.txt`.)

```
echo 1 | pc-spec_checker-kit.bat > result.txt
```

> Note: this saves text only, without colors or bar graphs. The simplest way to save is the "copy from the screen and paste into Notepad" method in 8-4.

15. File locations / Document locations

Files in this project

```
pc-spec_checker-kit/
├── pc-spec_checker-kit.bat    ← the actual tool (double-click this)
├── LICENSE                ← full license text (Apache 2.0)
├── NOTICE                 ← copyright notice
├── README.md              ← Korean intro (GitHub front page)
├── README.en.md           ← English intro
├── GUIDE.md               ← Korean complete guide
├── GUIDE.en.md            ← English complete guide (this document)
├── README.pdf / README.en.pdf    ← PDF versions of the intro
└── GUIDE.pdf  / GUIDE.en.pdf      ← PDF versions of the complete guide
```

> 📄 **The .md and .pdf have identical content.** Use `.md` for reading on screen, and `.pdf` for printing, sharing, or offline keeping.

If you can't find the downloaded file

- **Open the Downloads folder**: press `Windows key + E` (File Explorer) → click `Downloads` on the left.
- **Find what you just downloaded**: sort by `Date modified` so the newest is on top.
- **Search by name**: type `pc-spec_checker-kit` into the taskbar search box (the magnifier).

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16. Troubleshooting / Error handling

Find your symptom below.

① A blue "Windows protected your PC" window appears

→ This is normal. Click `More info` → `Run anyway`. (8-2)

② I double-clicked but the black window **flashed and closed instantly**

→ Usually one of these:

- You ran it **without extracting** (from inside the ZIP) → **Extract** the ZIP first, then run.
- Antivirus blocked it → check your antivirus quarantine/block list and trust it.
- To see for sure: open **PowerShell** in the file's folder and run it directly. (In the folder, `Shift + right-click` an empty area → `Open PowerShell window here` → type `.\pc-spec_checker-kit.bat` → Enter.) This keeps any error message visible so you can see the cause.

③ Text looks garbled (`□□□` or strange characters)

→ This tool outputs in English using UTF-8 (`chcp 65001`). It's usually fine, but old console settings may garble it. Running it in **Windows Terminal** (default on Windows 11) gives the cleanest result.

④ A program I definitely installed shows as `[X] not installed`

→ The tool checks **common install paths, your PATH, and the registry**. If you installed it in an unusual folder or it's not on PATH, it may not be found. Adding that program to your **system PATH** fixes detection. (The program itself still works fine.)

⑤ CPU temperature shows `N/A (admin required)`

→ Reading the temperature sensor needs **administrator rights**. Right-click the file → `Run as administrator` and it may appear. (Even as admin, some PCs have no supported sensor and show N/A — that's normal.)

⑥ The scan is very slow

→ The **Full Scan (1)** checks 193 tools and many sensors, so it can take **30–60 seconds**. Individual menus (e.g. 3 for CPU only) are much faster. In a hurry, pick only the items you need.

⑦ It doesn't work on Mac / Linux

→ This tool is **Windows only**. It does not run on Mac or Linux. (4. Prerequisites)

⑧ "Internet response time" shows `Slow` or nothing

→ Your internet is slow or disconnected. This is not a tool problem. Everything else works fine without internet.

⑨ Too many results — I can't scroll back up

→ Scroll up with the window's scroll bar or the mouse wheel. Or save to a file using "Save results to a text file" in 14. Commands and read it slowly.

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17. Frequently Asked Questions (FAQ)

Q. Is it safe to run? A. Yes. The tool **only reads info — it does not install, change, or send anything**. The source code is right inside the `.bat` file; open it with Notepad to verify. (20. Safety)

Q. Do I need administrator rights? A. No. Most things work without them. Only a few (like CPU temperature) show `N/A`.

Q. Does it work on Windows 10? A. Yes. Both Windows 10 and 11 are supported.

Q. Do I need internet? A. No. It works without it. With internet you only additionally see "internet response time."

Q. Where is my personal data sent? A. **Nowhere.** The only network action is a single connectivity ping (`8.8.8.8`), which sends no data.

Q. Does it leave any files or traces? A. No. The tool creates no files at all.

Q. What does "found in PATH" mean? A. The tool is installed but only its version number couldn't be read. It works fine.

Q. My antivirus warns it's dangerous. A. Because it's a `.bat` file, some antivirus reacts sensitively (false positive). You can read the code yourself, so trust it if needed.

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18. Glossary (plain language)

Term	Plain explanation
CPU	The computer's brain. Handles all calculations.
Core	The number of workers inside the CPU. More = more done at once.
Thread	A stream of work each core handles at once. Usually 2× the cores.
RAM	The computer's work desk. Wider = runs more programs at once.
DDR4 / DDR5	RAM generations. DDR5 is faster than DDR4.
GPU (graphics card)	Handles screen, games, video, and AI work.
VRAM	The graphics card's own memory. More = better for high-res work.
SSD	Fast storage. Quick boot and program launch.
HDD	Slow but high-capacity storage (hard disk).
NVMe	The fastest kind of SSD (much faster than a regular SSD).
BIOS	The first basic program that runs when the computer turns on.
Motherboard	The big board everything plugs into (the computer's body).
PATH	The computer's address book for finding programs. Registered ones run anywhere.
Driver	The software that makes a part work (e.g. graphics driver).
Antivirus	A security program that blocks malware.
Firewall	A security wall that blocks outside intrusion.
UAC	The security feature that asks "Allow?" when a program tries to change the system.
BitLocker	Windows' disk encryption feature.
WSL	A feature to run Linux inside Windows (for developers).
Git	A tool that manages code change history (a developer's "undo").
Node.js	A tool that runs JavaScript on your PC (essential for web dev).
Docker	A tool that runs programs in isolated environments.
IDE / editor	A specialized program for writing code (like VS Code).
CLI	Using a tool by typing commands (the black window).

Term	Plain explanation
.bat file	A Windows script file that runs commands automatically.
PowerShell	A powerful command tool built into Windows.
ping	A signal that checks if something responds ("are you there?").

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19. For people who code with AI

It's especially useful if you code with AI tools like ChatGPT, Claude, Cursor, or Copilot.

- **Check your environment**: AI told you to run `npm install` but it fails? Use **item 13 (Dev Tools)** to check if Node.js and npm are installed.
- **Tell the AI about your environment**: copy the full-scan (item 1) results and paste them to the AI, and it can give precise guidance tailored to your PC.
- **See what to install**: **item 15 (Score)** recommends the tools you're missing.

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20. Safety / Privacy

This tool does **NOT**:

- ❌ install anything.
- ❌ change any settings.
- ❌ create or delete any files.
- ❌ send your data anywhere.

All this tool does is:

- ✅ **read** your PC info using built-in Windows features
- ✅ **show** it in the window

> The only network action is a single connectivity **ping (`8.8.8.8`)**, and it sends no personal data. **The entire source code is inside the `.bat` file.** Open it with Notepad to verify.

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21. License / Copyright / Commercial use

> **Simple one-line summary**: Anyone may **use, modify, redistribute, and even use it commercially — for free**. Just **keep the copyright notice and a copy of the license**, and note it comes with **no warranty (use at your own risk)**.

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3. **State changes**: if you modified files, **prominently mark them as "changed."**
4. **Keep NOTICE**: include the attribution notices from the `NOTICE` file in your distribution.

🚫 Cautions / prohibitions

- **No trademark rights are granted.** You may not use names/logos like "SoDam AI Studio" or "PC Spec Checker" **as if they were your own product** (mentioning the origin is fine).
- **Patent clause**: a patent license from contributors is included, but if you **file a patent lawsuit** against this project, your granted patent rights terminate.

⚠️ Warranty / liability (disclaimer)

- This tool is provided **"AS IS"**, with **no warranty whatsoever** (including merchantability, fitness for a particular purpose, and non-infringement).
- The copyright holder and contributors are **not liable for any damages** arising from its use.
- Deciding whether and how to use it, and the associated risk, is **your own responsibility**.

🧩 Third-party components

- This tool does **not bundle or redistribute any third-party code/libraries**.
- At run time it uses only Windows built-in features (PowerShell, WMI/CIM, registry), which are already part of your Windows and governed by **Microsoft's license** (not redistributed by this project).

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22. Changelog

v3.0 (current)

- Expanded the menu from 9 to **15** items (+ item 0 to exit)
- Greatly expanded the dev-tools scan from ~50 to **193 tools** (17 categories)
- Added: Battery/Power, Security Status, Startup Programs, Audio/USB/Bluetooth, Installed Apps
- Added: Node version managers, Python ecosystem tools, AI coding tools, Cloud CLIs, Linters/Formatters
- Score system upgrade: 5 categories (was 3) + letter grades (S/A/B/C/D) + weakness analysis
- Better handling of tools that emit errors; improved wildcard resolution for version-specific install paths

v2.2

- Deep detection engine (PATH + known paths + registry)
- 50+ dev tools, score system (100 points, 3 categories)

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23. How to contribute

If you find a tool that isn't detected, or know a common install path that's missing, contributions are welcome.

- Open an **Issue** (problem/suggestion) or a **Pull Request** (proposed edit) on the GitHub repository.
- Contributions are considered to be provided under the same **Apache License 2.0** as this project.

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24. Contact / Support

- Project: **PC Spec Checker** (by SoDam AI Studio)
- Repository: `https://github.com/sodam-ai/pc-spec_checker-kit`
- Questions / bug reports: use the **Issues** tab on the repository above.

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